

DUAL MONTE CARLO UNCERTAINTY FRAMEWORK FOR NEURAL NETWORKS: APPLICATION TO A QUANTUM VOLTAGE STANDARD

G. Schneider^(1,2), R. Iuzzolino⁽¹⁾, M. Real^(1,2), M. Arifovic⁽³⁾ y G. Maculotti⁽⁴⁾

(1) Instituto Nacional de Tecnología Industrial (INTI), (2) Instituto de Calidad Industrial, Universidad Nacional de San Martín (INCALIN-UNSAM), (3) TÜBİTAK Ulusal Metroloji Enstitüsü, Gebze-Kocaeli, Türkiye (4) DIGEP, Politecnico di Torino, Turin, Italy

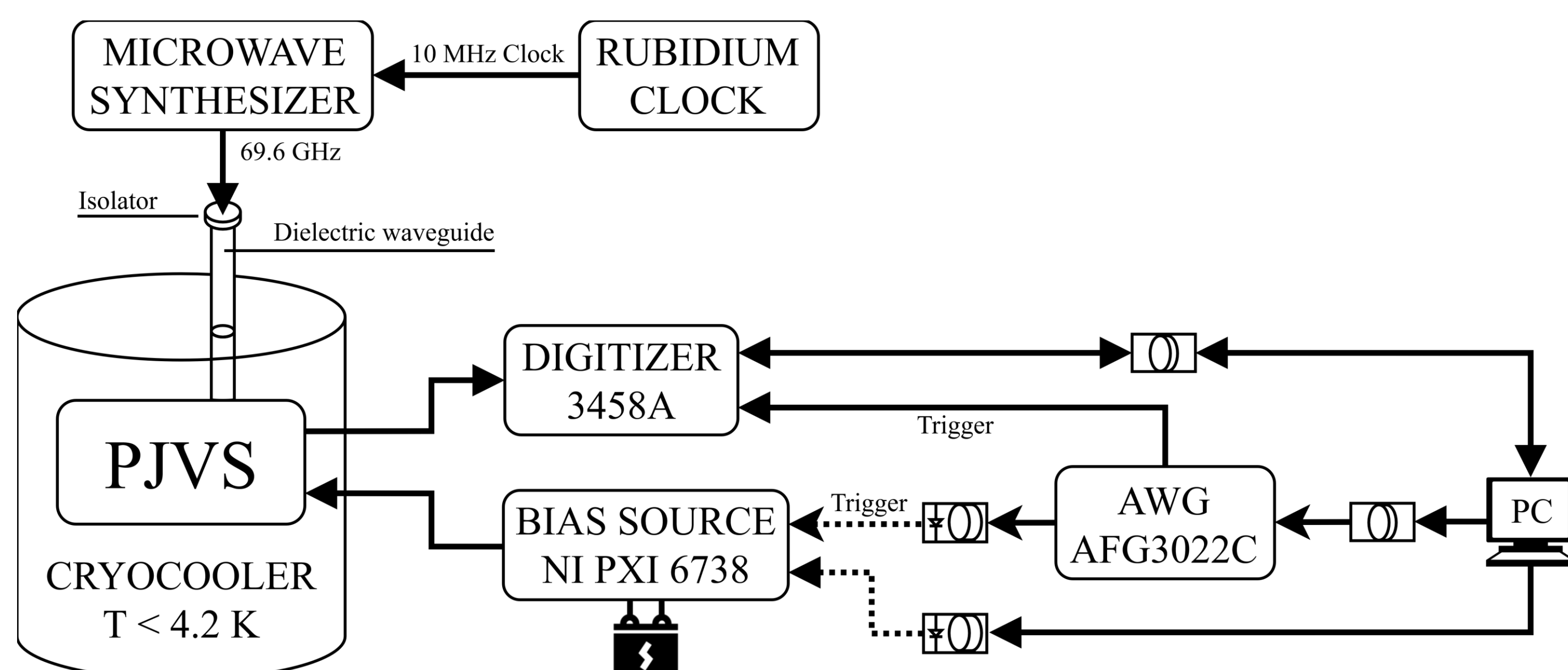
INTI, Av. Gral. Paz 5445, San Martín, Bs. As., Argentina | gschneider@inti.gob.ar

1. Introduction

Programmable Josephson Voltage Standards (PJVS) [1] enable traceable DC voltage realization with extremely low uncertainty. Neural-network-based digital twins can model PJVS behavior for prediction and optimization, but their uncertainty contributions must be quantified in compliance with the GUM [2] for metrological use. This work proposes the Dual Monte Carlo Uncertainty Framework (DMCUF), which combines GUM-compliant Monte Carlo [3] propagation of measurement uncertainties with Monte Carlo Dropout [4] —to estimate model uncertainty— and bias uncertainty evaluation from reference measurements. The method is demonstrated using DC voltage measurements from a PJVS, part of a neural-network-based digital twin developed within the scope of the EURAMET ViDiT project [5] to optimize the system and reduce setup and measurement times.

2. Method

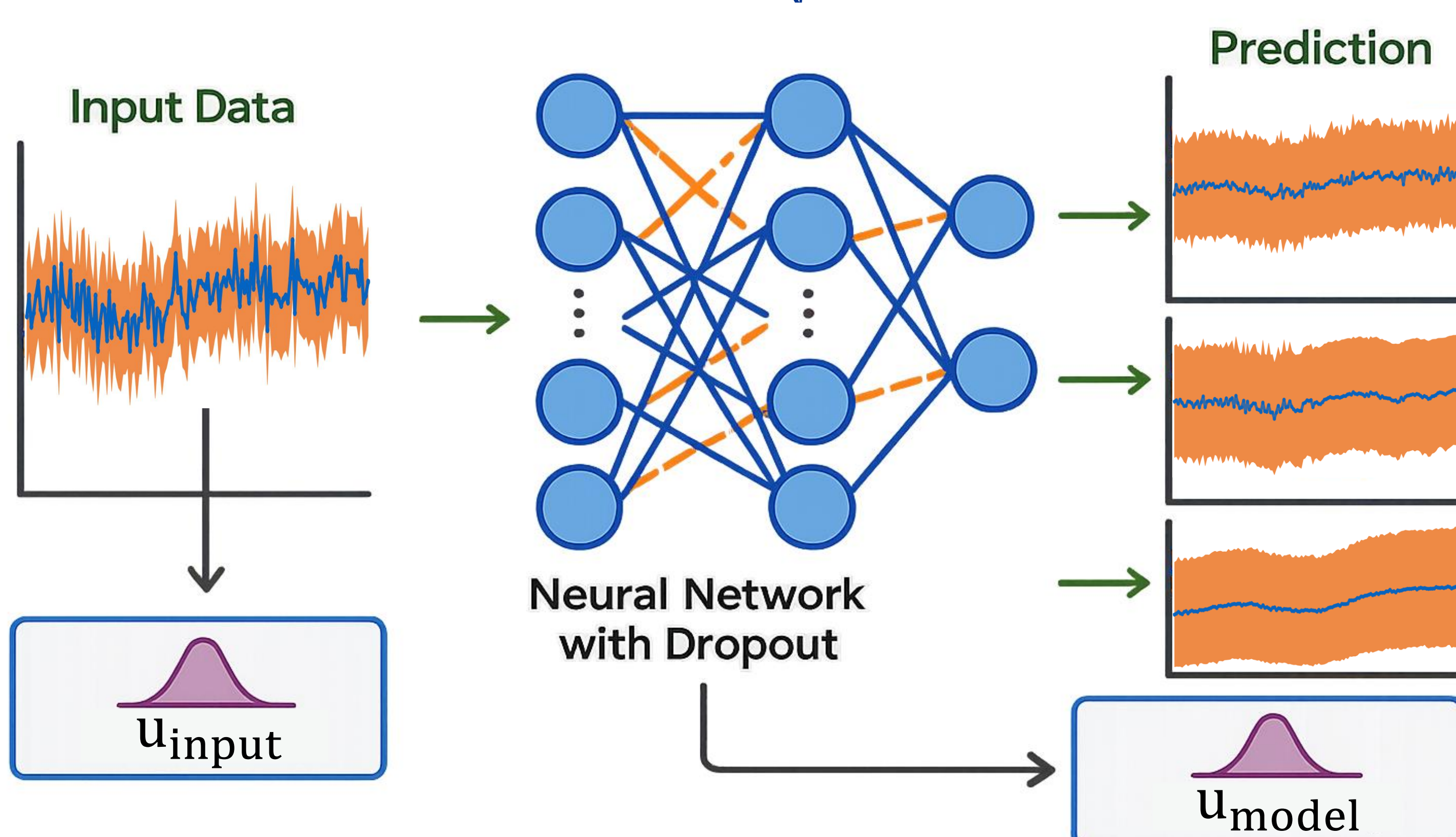
The datasets were generated using measurements of a DC voltage step of 1.0001086679 V, from the TÜBİTAK UME 10 V PJVS system, with the following experimental setup:



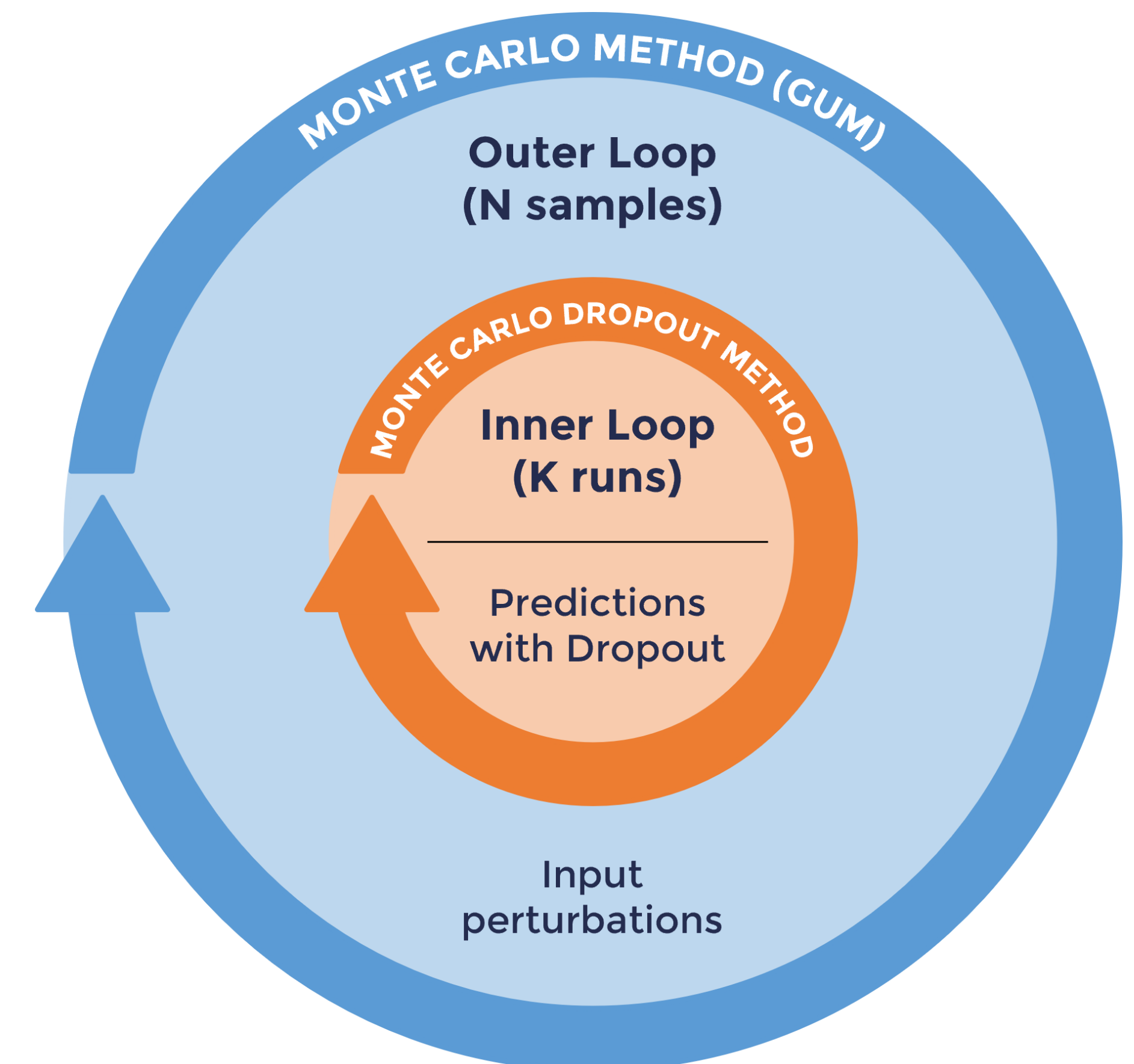
Let $\hat{y} = f_{\omega}(x; m) + b$ denote the predictions of a trained neural network, where x is the vector of inputs that were obtained from the DC step measurements of the PJVS, ω denotes a fixed snapshot of trained weights, m is a stochastic dropout mask, and $b = \langle \hat{y}_{\text{ref}} - y_{\text{ref}} \rangle$ is the bias of model's predictions obtained as the mean of the residuals between reference measurements y_{ref} and the predictions $\hat{y}_{\text{ref}}$, from where we can obtain the model bias uncertainty u_b . The total variance of $\hat{y}$ is decomposed using the law of total variance, $\text{Var}(\hat{y}) = E_x[\text{Var}(\hat{y} | x)] + \text{Var}_x(E[\hat{y} | x])$, and the uncertainty components are:

$$u_{\text{input}} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\bar{y}_i - \hat{\mu})^2}, \quad u_{\text{model}} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{1}{K-1} \sum_{j=1}^K (y_{i,j} - \bar{y}_i)^2 \right)},$$

resulting in a combined uncertainty: $u_c = \sqrt{u_{\text{input}}^2 + u_{\text{model}}^2 + u_b^2}$.



Monte Carlo Dropout method with input data uncertainty



Representation of the double loop uncertainty evaluation method

3. Results

Three architectures were tested as surrogate models of the PJVS digital twin: a multilayer perceptron (MLP), a long short-term memory network (LSTM), and a convolutional neural network (CNN). All models were trained separating in 70% training, 20% validation, and 10% test in a one-step-ahead sequence-to-sequence setting. The DMCUF used $N = 1000$ Monte Carlo input samples with an uncertainty $\sigma_{\text{input}} = 79$ nV estimated from the measurement series, and $K = 200$ Monte Carlo Dropout realizations (dropout rate = 40%). Coverage factor is $k = 1$, and 95% empirical coverage is approximated by the interval $[\text{mean} \pm 2 u_c]$.

Comparison of predictive performance (test set)

Model	RMSE (nV)	Bias (nV)	Coverage 95%
MLP	64	-2	96.47%
LSTM	66	-13	90.93%
CNN	69	-17	73.05%

Comparison of uncertainty components (test set)

Model	u_b (nV)	$\langle u_{\text{input}} \rangle$ (nV)	$\langle u_{\text{model}} \rangle$ (nV)	$\langle u_c \rangle$ (nV)
MLP	3	22	67	71
LSTM	3	16	50	53
CNN	3	9	33	35

4. Conclusions

DMCUF provides a GUM-compliant uncertainty evaluation method for neural-network-based models, as demonstrated in this PJVS digital twin application. The method provides a transparent and reproducible way to quantify uncertainty contributions from both inputs and model structure. Results showed that combined uncertainty is strongly dependent on model architecture and dropout layer configuration. Due to its superior predictive performance, the MLP model proved to be the one that best represents the measured data. The approach presented is easily applicable to other metrological systems that employ data-driven models.

5. References

- [1] S. Bauer et al. Measurement Science and Technology, 2022.
- [2] BIPM, IEC, IFCC, ILAC, ISO, IUPAC, IUPAP, OIML. GUM, JCGM 100:2008.
- [3] BIPM, IEC, IFCC, ILAC, ISO, IUPAC, IUPAP, OIML. GUM Supplement 1, JCGM 101:2008.
- [4] Y. Gal and Z. Ghahramani. 33rd International Conference on Machine Learning, 2016.
- [5] ViDiT Project, "22dit01 vidiT: Trustworthy virtual experiments and digital twins, 2024.